

Kamil Mikolaj

Machine Learning Engineer

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📍 Copenhagen, Denmark

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Summary

ML engineer with a track record of owning systems end-to-end across MLOps, computer vision, and 3D reconstruction. Comfortable at every layer of the stack — from C++ to Kubernetes-based retraining pipelines. Background in a domain where evaluation rigour and data quality aren't optional, which shapes how I approach production ML generally.

Experience

Dalux, Machine Learning and Computer Vision Engineer

Copenhagen

Applying ML and computer vision to BIM and construction site understanding — building production systems across data pipelines, model training, and inference, with visibility into cloud deployment (AWS, Kubernetes, Argo) and monitoring (Prometheus/Grafana).

Nov 2025 – present
6 months

GE Healthcare, Visiting Researcher

Herlev

Research and optimization of motion-compensated 3D ultrasound reconstruction for laparoscopic liver workflows.

Jan 2025 – Mar 2025
2 months

- Targeted sub-minute runtime per sequence.

Technical University of Denmark (DTU), PhD Student

Kongens Lyngby

End-to-end ML/CV research in obstetric ultrasound in collaboration with Rigshospitalet.

Aug 2022 – Oct 2025
3 years 3 months

- Technical lead for 3D reconstruction, fetal-growth modeling, heart segmentation, and analysis.
- Cut training time by 3x at matched quality for a 3D ultrasound reconstruction pipeline. [Publication](#)
- Built an automated large-scale ultrasound pipeline for image-based fetal growth regression.
- Reduced measurement error by 13% versus current clinical practice. [Publication](#)
- Contributed to two UK patents. [Granted](#), [Pending](#)
- Built a heart segmentation model with end-user input and deployed a real-time tablet prototype.

SPS - Specialpaedagogisk Stotte, Pedagogical Support Teacher, Part-time

Kongens Lyngby

Providing academic and mentoring support to students with special educational needs.

Apr 2024 – Dec 2025
1 year 9 months

Copenhagen Academy for Medical Education and Simulation (CAMES), Research

Copenhagen

Assistant, Part-time

June 2021 – June 2022

Collaborative work between DTU and Rigshospitalet on ML in obstetric ultrasound.

1 year 1 month

- Contributed to data preprocessing, fetal growth prediction, and multi-institutional project coordination.

Education

PhD **Technical University of Denmark (DTU)**, Applied Mathematics and Computer Science (Visual Computing)

Kongens Lyngby

Aug 2022 – Oct 2025

- Thesis: From Automated 2D Analysis to 3D Reconstruction: Advancing Ultrasound Imaging with AI-Driven Methods. [Download](#)
- [Courses](#)

MSc **Technical University of Denmark (DTU)**, Autonomous Systems Engineering (Honours Programme)

Kongens Lyngby

Sept 2020 – June 2022

- Thesis: Automatic Detection of Anatomical Structures in Fetal Hearts.
- [Courses](#)

Skills

Modelling: Deep Learning, Segmentation, 3D Reconstruction, NeRF, Semi-supervised Learning, Unsupervised Learning, Classification

Engineering: Python, C/C++, Systems Design, Inference Optimization, API Design

Infrastructure: MLOps, Cloud Deployment, Workflow Orchestration, Monitoring, Containerization

Data: Pipeline Design, Annotation, Dataset Curation, Data Quality Assurance

Computer Vision: Image Processing, Image Analysis, 3D Reconstruction

Tools

ML Frameworks: PyTorch, OpenCV, scikit-learn

Infrastructure & Cloud: AWS, Kubernetes, Docker, Argo Workflows, Argo Rollouts

Monitoring & Observability: Prometheus, Grafana, Loki

MLOps: MLflow, ONNX, Helm

Languages: Python, C/C++, MATLAB, LaTeX

Awards

UK Patent Application GB2636226

2023

A method of, and apparatus for, improved estimation of fetal characteristics.

UK Patent Application GB2636227

2023

An improved method of, and apparatus for, ultrasound examination to extract fetal characteristics.